



KPR INSTITUTE OF ENGINEERING AND TECHNOLOGY

(Autonomous Institution, Affiliated to Anna University, Chennai)

Avinashi Road, Arasur, Coimbatore - 641 407.

Assignment – II

Course Code and Title	U25CSG05 – Computational Problem Solving using Python		
Year & Semester	I & 02		
Degree & Branch	B.Tech & CSE(AIML)		
Regulation	R2025	Max. Marks:	20

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Section: A

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1. Design a Python program for a Student Management System: (2)CO3

- Use a class Student with attributes name, roll_no, and marks.
- Create objects for 3 students and display their details.

Ans:

class Student:

```
    """Class to represent a Student with name, roll number, and marks."""
```

```
    def __init__(self, name, roll_no, marks):
```

```
        self.name = name
```

```
        self.roll_no = roll_no
```

```
        self.marks = marks
```

```
    def display_details(self):
```

```
        """Method to display student details."""
```

```
        print(f"Name: {self.name}")
```

```
        print(f"Roll Number: {self.roll_no}")
```

```
        print(f"Marks: {self.marks}")
```

```
        print("-" * 20)
```

```
# Creating objects for 3 students
```

```
student1 = Student("Alice", 101, 85)
```

```
student2 = Student("Bob", 102, 92)
```

```
student3 = Student("Charlie", 103, 78)
```

```
# Displaying their details
```

```
print("Student Management System Records:\n")
```

```
student1.display_details()
```

```
student2.display_details()
```

```
student3.display_details()
```

Output:

```
Student Management System Records:
```

```
Name: Alice
```

```
Roll Number: 101
```

```
Marks: 85
```

```
-----
```

```
Name: Bob
```

```
Roll Number: 102
```

```
Marks: 92
```

```
-----
```

```
Name: Charlie
```

```
Roll Number: 103
```

```
Marks: 78
```

```
-----
```

**2. Implement a Python program using list comprehension and sets:
(2)CO3**

- **Generate a list of numbers from 1 to 20.**
- **Extract even numbers and store them in a set.**
- **Display the final output.**

Ans:

```
numbers_list = [x for x in range(1, 21)]
```

```
even_numbers_set = {x for x in numbers_list if x % 2 == 0}
```

```
print("Original List:", numbers_list)
```

```
print("Even Numbers Set:", even_numbers_set)
```

Output:

```
===== RESTART: C:/Users/ADMIN/OneDrive/Desktop/New folder/even.py =====  
Original List: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20]  
Even Numbers Set: {2, 4, 6, 8, 10, 12, 14, 16, 18, 20}  
|
```

3. Develop a Python program to perform file handling operations (2)CO4

- **Write student details into a file.**
- **Read and display the stored content.**

Ans:

```
def write_student_details(filename):
```

```
    try:
```

```
        with open(filename, "w") as file:
```

```
            n = int(input("Enter the number of students: "))
```

```
            for i in range(n):
```

```
                print(f"\nEnter details for Student {i+1}:")
```

```
                roll_no = input("Roll No: ")
```

```
                name = input("Name: ")
```

```
                marks = input("Marks: ")
```

```
                record = f"{roll_no}, {name}, {marks}\n"
```

```
                file.write(record)
```

```
            print("\nStudent details successfully written to the file.")
```

```
        except Exception as e:
```

```
            print(f"An error occurred while writing: {e}")
```

```
def read_student_details(filename):
```

```

print("\n--- Stored Student Details ---")

try:
    with open(filename, "r") as file:
        content = file.read()
        if content:
            print(content)
        else:
            print("The file is empty.")
except FileNotFoundError:
    print("Error: The file was not found.")
except Exception as e:
    print(f"An error occurred while reading: {e}")

file_name = "student_records.txt"
write_student_details(file_name)
read_student_details(file_name)

```

Output:

```

> ===== RESTART: C:/Users/CCL/Desktop/python/student.py =====
Enter the number of students: 4

Enter details for Student 1:
Roll No: 24
Name: dharunya.S
Marks: 550

Enter details for Student 2:
Roll No: 25
Name: jothika.M
Marks: 550

Enter details for Student 3:
Roll No: 26
Name: kavi
Marks: 550

Enter details for Student 4:
Roll No: 31
Name: megha
Marks: 550

Student details successfully written to the file.

--- Stored Student Details ---
24, dharunya.S, 550
25, jothika.M, 550
26, kavi, 550
31, megha, 550
> |

```

4. Implement a Python program to process CSV data: (2)CO4

- Read employee details from a CSV file.
- Display name and salary of employees.

Ans:

```
import csv

with open('employees.csv', 'w', newline='') as file:

    writer = csv.writer(file)

    writer.writerow(["Name", "Department", "Salary"])

    writer.writerow(["John Doe", "IT", "75000"])

    writer.writerow(["Jane Smith", "HR", "62000"])

with open('employees.csv', mode='r') as file:

    reader = csv.DictReader(file)

    print(f"{'Name':<15}{'Salary':<10}")

    print("-" * 25)

    for row in reader:

        print(f"{'row['Name']':<15}{'row['Salary']':<10}")
```

Output:

```
===== RESTART: C:/Users/ADMIN/OneDrive/Desktop/New folder/csv file.py
Name           Salary
-----
Kavisharmitha   75000
jothika         62000
> |
```

5. Design a command-line calculator program: (2)CO4

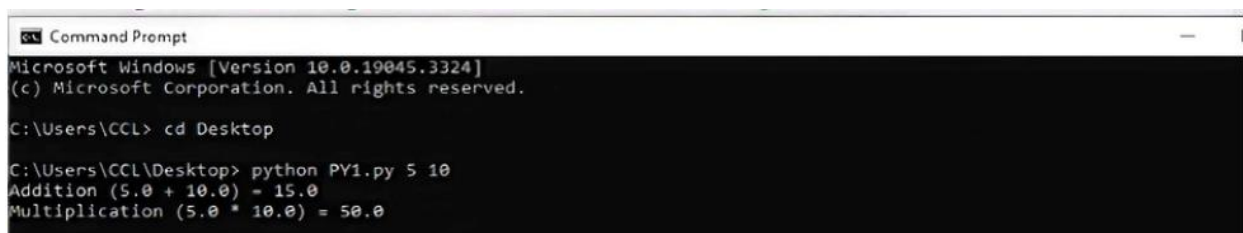
- Accept inputs using command line arguments.
- Perform addition and multiplication.

Ans:

```
import sys

if len(sys.argv) != 3:
    print("Usage: python script.py <number1> <number2>")
else:
    try:
        num1 = float(sys.argv[1])
        num2 = float(sys.argv[2])
        print(f"Addition ({num1}) + ({num2}) = {num1 + num2}")
        print(f"Multiplication ({num1} * {num2}) = {num1 * num2}")
    except ValueError:
        print("Please provide valid numbers as arguments.")
```

Output:



```
Command Prompt
Microsoft Windows [Version 10.0.19045.3324]
(c) Microsoft Corporation. All rights reserved.

C:\Users\CCL> cd Desktop

C:\Users\CCL\Desktop> python PY1.py 5 10
Addition (5.0 + 10.0) = 15.0
Multiplication (5.0 * 10.0) = 50.0
```

6. Develop a Python program to create a log file system: (2)CO4

- Store user activities in a file.
- Retrieve and display the logs.

Ans:

```
import datetime

def log_activity(activity):
    """Stores user activities in a log file."""
    timestamp = datetime.datetime.now().strftime("%Y-%m-%d%H:%M:%S")
    with open("system.log", "a") as file:
        file.write(f"[{timestamp}] - {activity}\n")
log_activity("User Admin logged in")
log_activity("File 'data.csv' uploaded")
print("--- System Logs ---")
with open("system.log", "r") as file:
    print(file.read())
```

Output:

```
--- System Logs ---
[2026-04-05 14:26:21] - User Admin logged in
[2026-04-05 14:26:21] - File 'data.csv' uploaded
|
```

7. Develop a Python program using NumPy arrays: (2)CO5

- Create a 2D array.
- Perform slicing and display results.

Ans:

```
import numpy as np

matrix = np.array([
    [10, 20, 30, 40],
    [50, 60, 70, 80],
    [90, 100, 110, 120]
])

print("Original 2D Array:\n", matrix)

sliced_matrix = matrix[0:2, 1:3]

print("\nSliced Array (Rows 0-1, Cols 1-2):\n", sliced_matrix)
```

Output:

```
----- RESTART: C:/Users/ADMIN/C
Original 2D Array:
[[ 10  20  30  40]
 [ 50  60  70  80]
 [ 90 100 110 120]]

Sliced Array (Rows 0-1, Cols 1-2):
[[20 30]
 [60 70]]
```

8. Implement a Python program using Pandas DataFrame: (2)CO5

- Create a dataset with student marks.
- Filter students scoring above average.

Ans:

```
import pandas as pd

data = {
    'Student': ['Alice', 'Bob', 'Charlie', 'David', 'Eve'],
    'Marks': [88, 45, 92, 75, 60]
}

df = pd.DataFrame(data)

average_mark = df['Marks'].mean()

print(f"Class Average: {average_mark}\n")

above_average_students = df[df['Marks'] > average_mark]

print("Students Scoring Above Average:")

print(above_average_students)
```

Output:

```
===== RESTART: C:/Users/ADMIN
Class Average: 72.0

Students Scoring Above Average:
   Student  Marks
0    Alice     88
2   Charlie     92
3    David     75
```

9. Develop a Python program to perform data visualization: (2)CO5

- Plot a histogram using Matplotlib.
- Display and interpret the output.

Ans:

```
import matplotlib.pyplot as plt

scores = [45, 55, 50, 65, 75, 85, 90, 45, 60, 55, 70, 75, 80, 95]

bins = [40, 50, 60, 70, 80, 90, 100]

plt.hist(scores, bins=bins, color='skyblue', edgecolor='black')

plt.title("Student Score Distribution")

plt.xlabel("Score Range")

plt.ylabel("Number of Students")

plt.grid(axis='y', alpha=0.75)

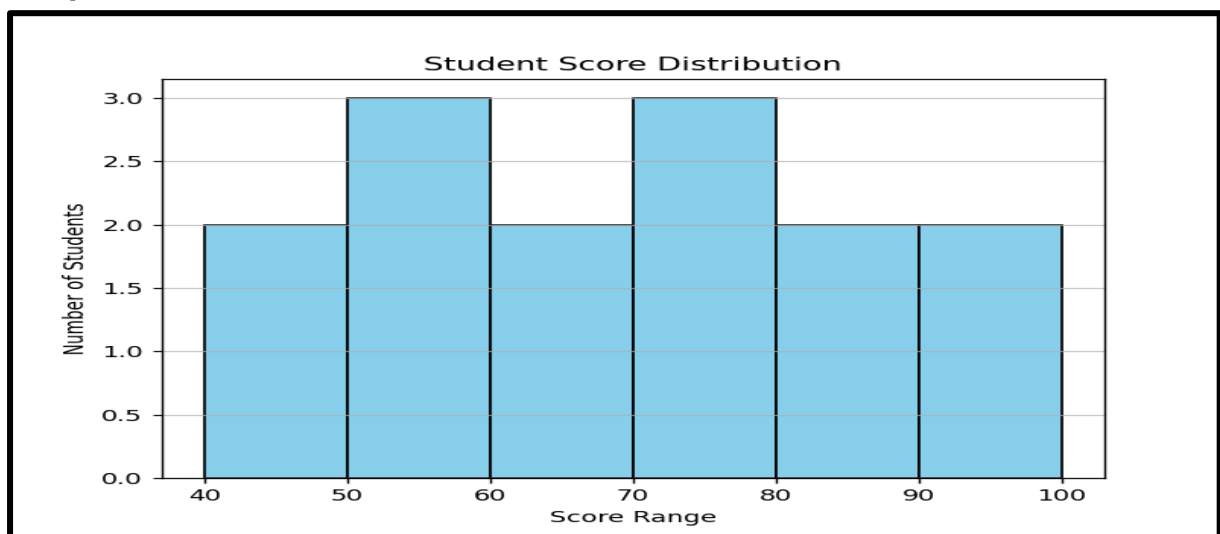
plt.show()

print("Interpretation: The histogram shows the distribution of student scores. ")
```

"The highest concentration of scores falls in the 50-60 and 70-80 bins, "

"indicating where the majority of the class performed.")

Output:



10. Design a Python program for data cleaning: (2)CO5

- Identify missing values in dataset.
- Replace them with suitable values.

Ans:

```
import pandas as pd
import numpy as np
data = {
    'Name': ['Karthik', 'Lakshmi', 'Murali', 'Nisha', 'Om'],
    'Age': [20, np.nan, 21, 22, np.nan],
    'Score': [85, 90, np.nan, 88, 76]
}
df = pd.DataFrame(data)
print("--- Original Dataset with Missing Values ---")
print(df)
print("\n--- Missing Values Count ---")
print(df.isnull().sum())
df['Age'] = df['Age'].fillna(df['Age'].median())
df['Score'] = df['Score'].fillna(df['Score'].mean())
print("\n--- Cleaned Dataset ---")
print(df)
```

Output:

```
REGDIR: C:/Users/ADMIN/Desktop/Python/
--- Original Dataset with Missing Values ---
   Name  Age  Score
0  Karthik 20.0  85.0
1  Lakshmi  NaN  90.0
2   Murali 21.0   NaN
3   Nisha 22.0  88.0
4     Om  NaN  76.0

--- Missing Values Count ---
Name      0
Age       2
Score     1
dtype: int64

--- Cleaned Dataset ---
   Name  Age  Score
0  Karthik 20.0  85.00
1  Lakshmi 21.0  90.00
2   Murali 21.0  84.75
3   Nisha 22.0  88.00
4     Om 21.0  76.00
```